

Kumar Sahab

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Dear Hiring Manager,

I am writing to express my strong interest in the AI/ML role at your company.

With a solid foundation in software engineering and a deep technical background in artificial intelligence, I bring over **seven years** of industry experience, **three years** in **full-stack development** and **four years** in **AI/ML engineering**.

I embarked on my software development journey at **Cognizant** as a **full stack developer**, where I played a pivotal role in developing a sophisticated web application for a leading American insurance company. This experience enriched my understanding of software design and development principles, OOPs, and design patterns, which are crucial for building scalable and sustainable software solutions.

My contributions included the end-to-end development of **two web applications**, implementation of over 20 significant enhancements and new features, which markedly improved the user experience. Additionally, I addressed over 50 critical and high-priority technical issues, effectively mitigating production challenges at scale. My efforts also extended to the technological advancement of more than 10 web applications.

In the subsequent phase of my career, I was privileged to join the team at **Samsung Research** as a **Senior Machine Learning Engineer**. My tenure was characterized by an in-depth exploration of avant-garde AI technologies, with a particular emphasis on **natural language processing (NLP)**, **computer vision** and **Generative AI**. I was entrusted with the comprehensive development and deployment of deep learning solutions, a responsibility that spanned the entire gamut of tasks from data collection, preparation, cleaning, and preprocessing to model development, training, compression, and deployment.

While at **Samsung**, I successfully developed three end-to-end models utilizing **transformers** and **CNN** architectures from the ground up. Additionally, I played an instrumental role in the creation of two other deep learning models and the establishment of an on-device model training pipeline using Open Neural Network Exchange (**ONNX**), furthering the frontier of AI innovation. I also contributed to an **LLM-powered** device security status feature that interpreted system flags and permission signals to generate intuitive device health summaries using on-device **generative AI**, improving the security user experience while preserving user data privacy. These efforts contributed to the AI features on flagship Samsung smartphones and earned me **two Samsung Excellence Awards** and **3 times recognition from peers as a technical expert**. Additionally, I achieved **Samsung SWC Professional & Workera AI Expert** certification here.

Currently, I serve as a **Senior Data Scientist** at **Aarki**, an **advertising** technology startup specializing in **real-time bidding**. I lead the development and optimization of machine learning models across the bidding pipeline to ensure competitive pricing while maximizing user acquisition. I architected high-impact **install propensity** models that delivered a **1.5x performance lift** over the champion model and scaled them to production with **~\$2K in daily spend**, generating approximately **\$600 per day** in incremental revenue through improved audience targeting. I also developed a **Transformer-based lifetime value (LTV) model** to predict 7-day purchase intent using only the first 24 hours of post-install activity, reducing the label maturity window from seven days to one and significantly accelerating model iteration cycles. In addition, I conducted end-to-end bid funnel analyses to identify and resolve critical system bottlenecks across multiple stages of the bidding process. My work spans large-scale data analysis, feature engineering, experimentation, calibration, deployment, and monitoring, and I engineered robust training and deployment pipelines with automated workflows using Prefect to improve reliability and velocity.

I am excited about the opportunity to contribute to your company's mission. I am looking forward to discussing how my background, skills, and enthusiasms align with the goals of your company. I am confident in my ability to make meaningful contributions to your team and am excited about the prospect of showcasing my capabilities.

Warm Regards,

Kumar Sahab